Strategic reshaping of the attentional axis in primate superior colliculus

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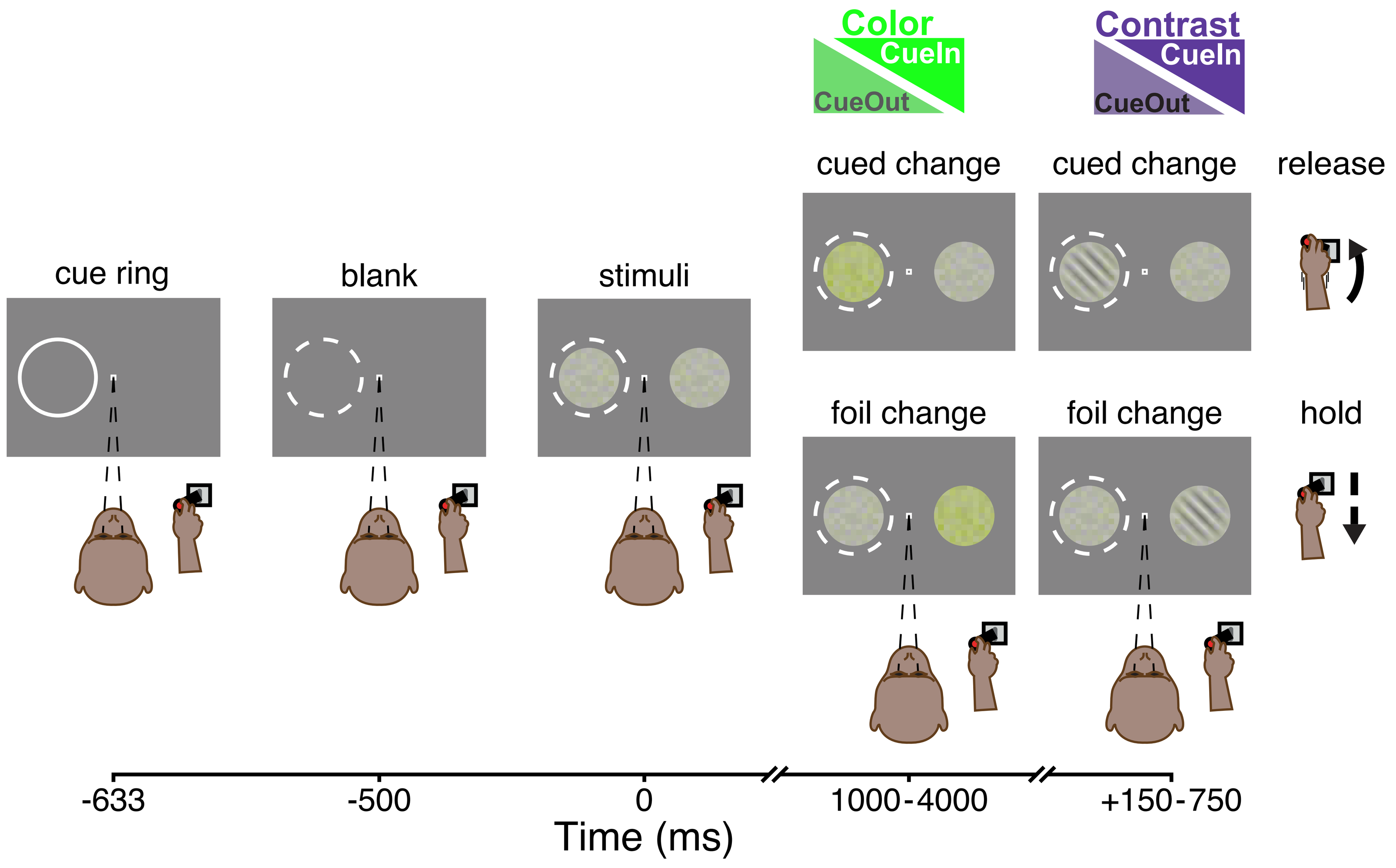
Introduction

- Visual attention adapts to perceptual demands — deployed focally when stimuli are ambiguous, relaxed when they are clear.
- In laboratory studies of attention, stimulus conditions are fixed, removing natural variation that might recruit distinct neuronal modulation policies.
- Accordingly, attention-related modulation is usually characterized as low-dimensional gain, but this may reflect the structure of the paradigms rather than the mechanism.
- Does the attention modulation axis reconfigure when task demands change, as they do in natural vision?

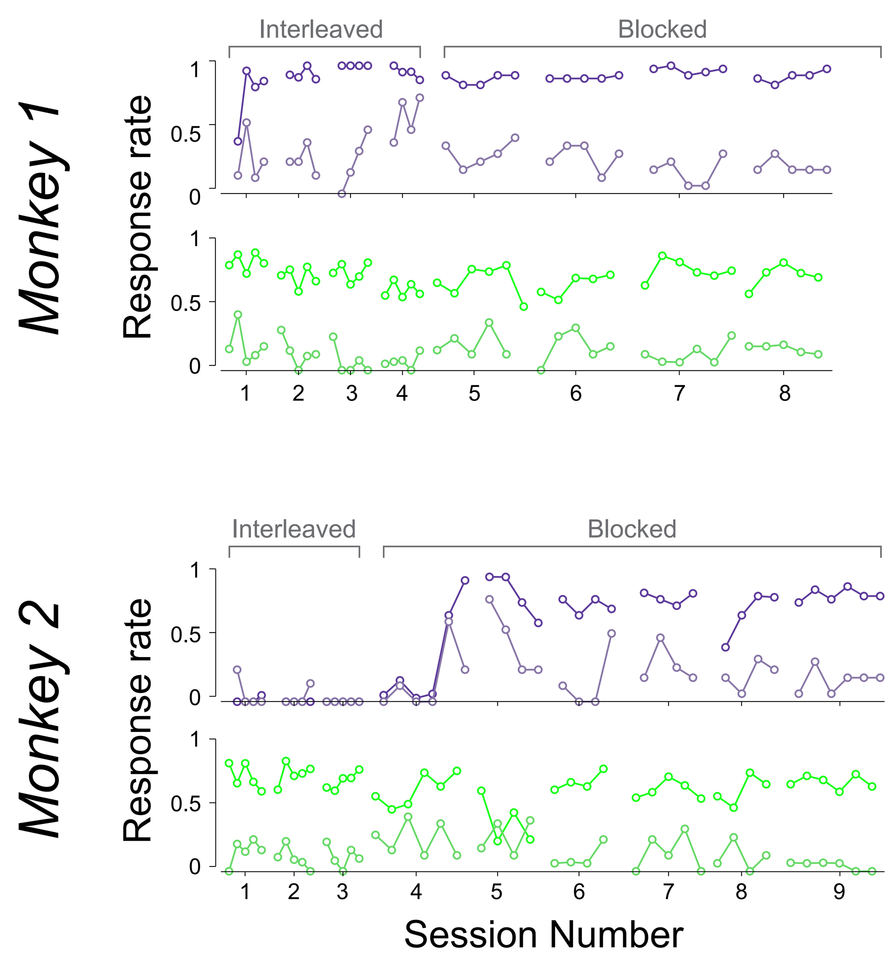
Conclusions

- Predictable task demands produce change-type-specific preparatory states in SC that involve geometric reconfiguration of the attentional axis and not only gain scaling up or down.
- Preparatory states occupy a shared subspace; reconfiguration is a rotation within that subspace.
- The apparently low-dimensional character of attentional modulation may reflect the constraints of standard paradigms rather than the mechanism itself.
- Does the preparatory state predict behavioral outcomes? Future work must functionally test the relevance of this neuronal population state for behavior.

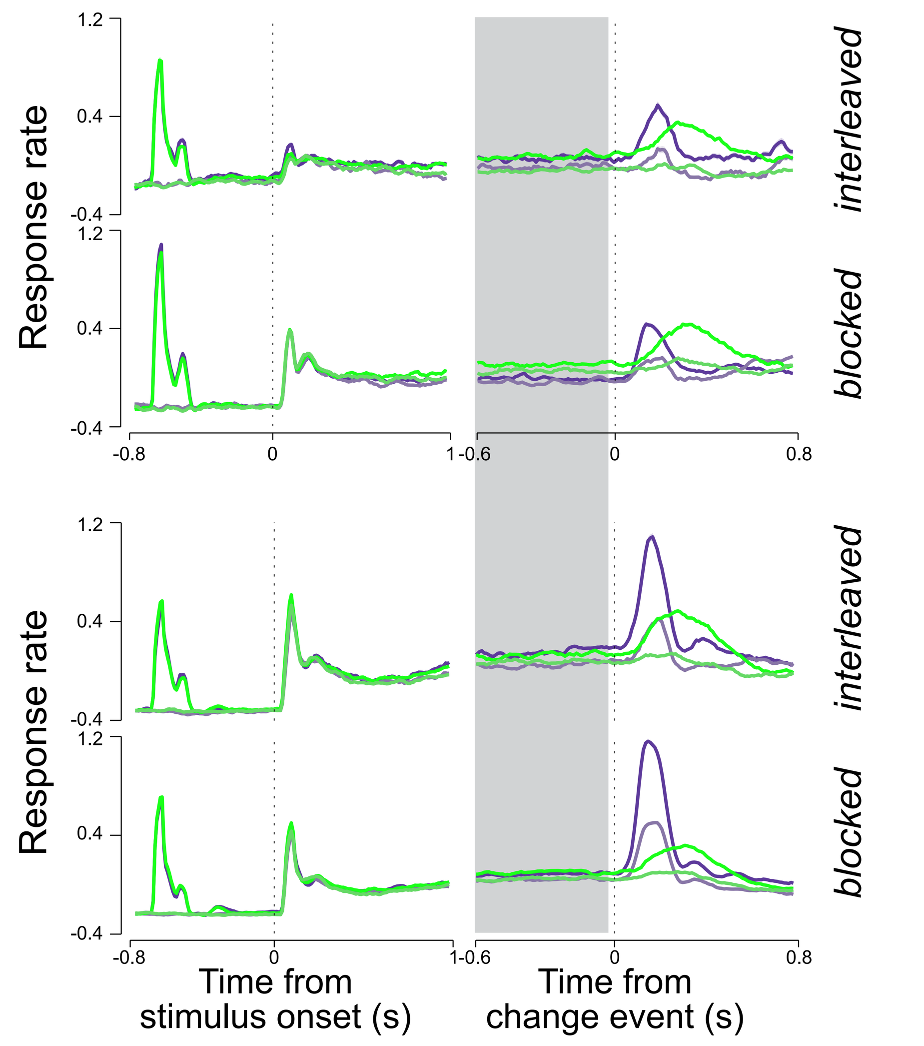
a. Covert attention task with two distinct feature changes



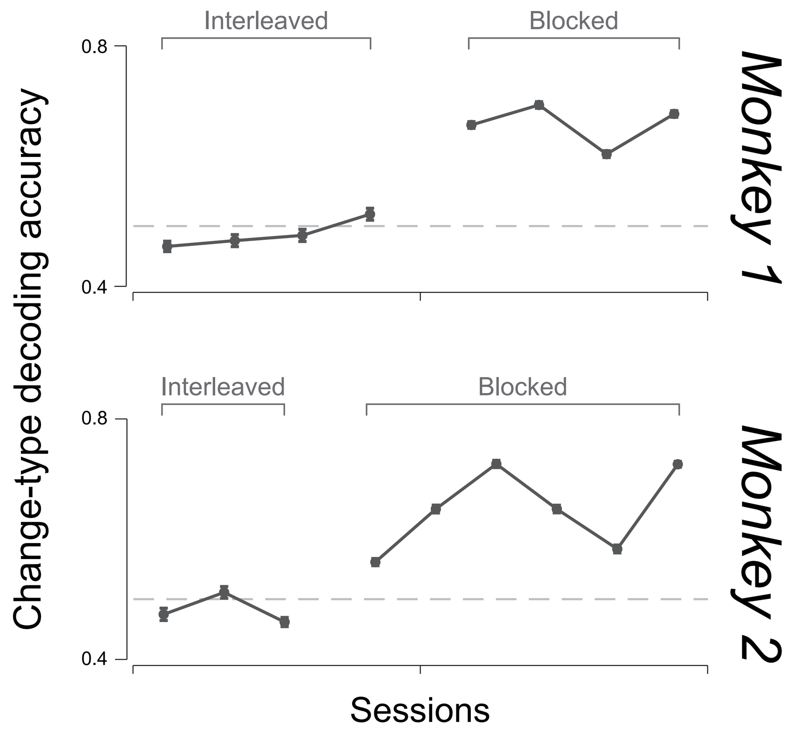
b. Task performance



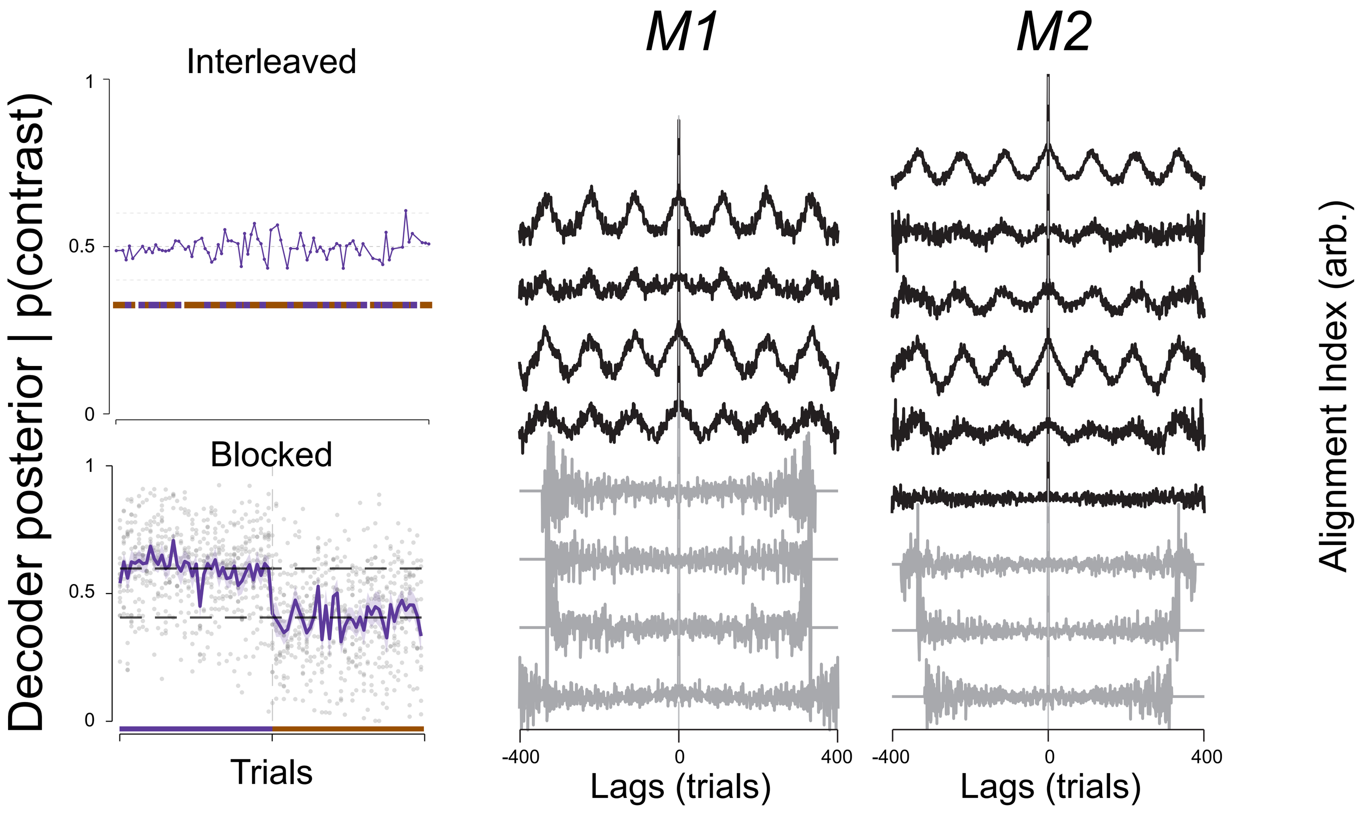
c. Blocked presentation of feature changes results in a "preparatory state" of activity



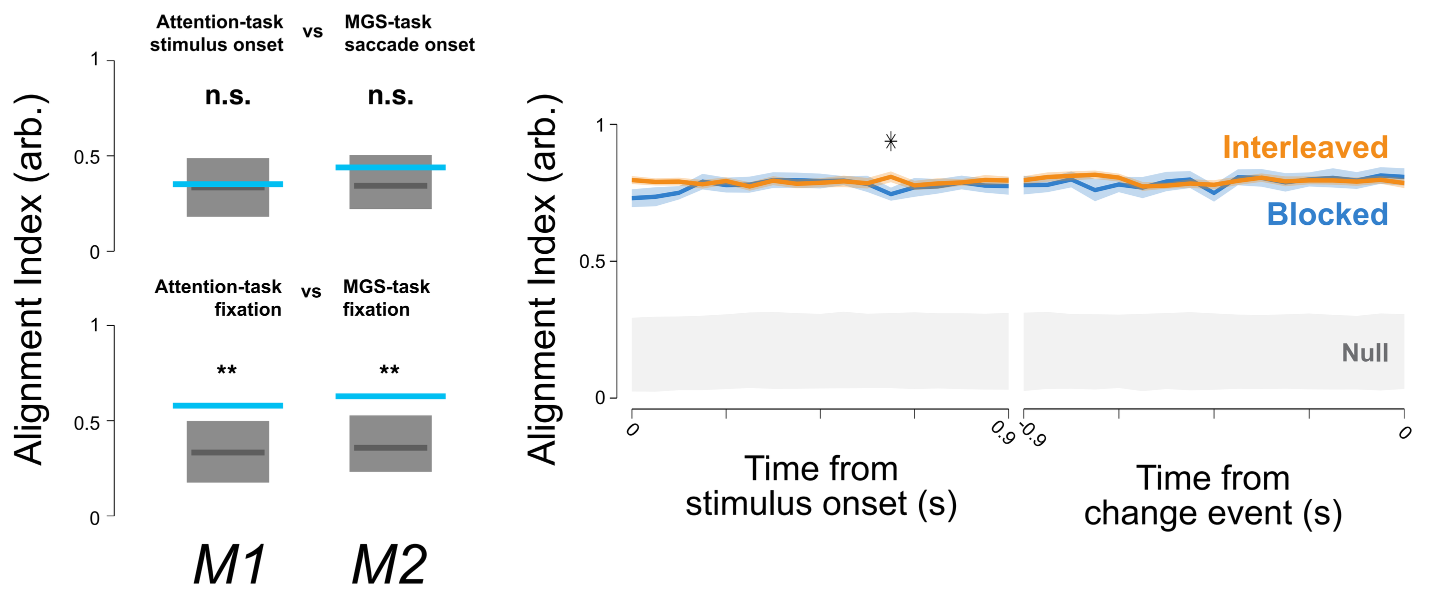
d. Predictable structure drives distinct preparatory states



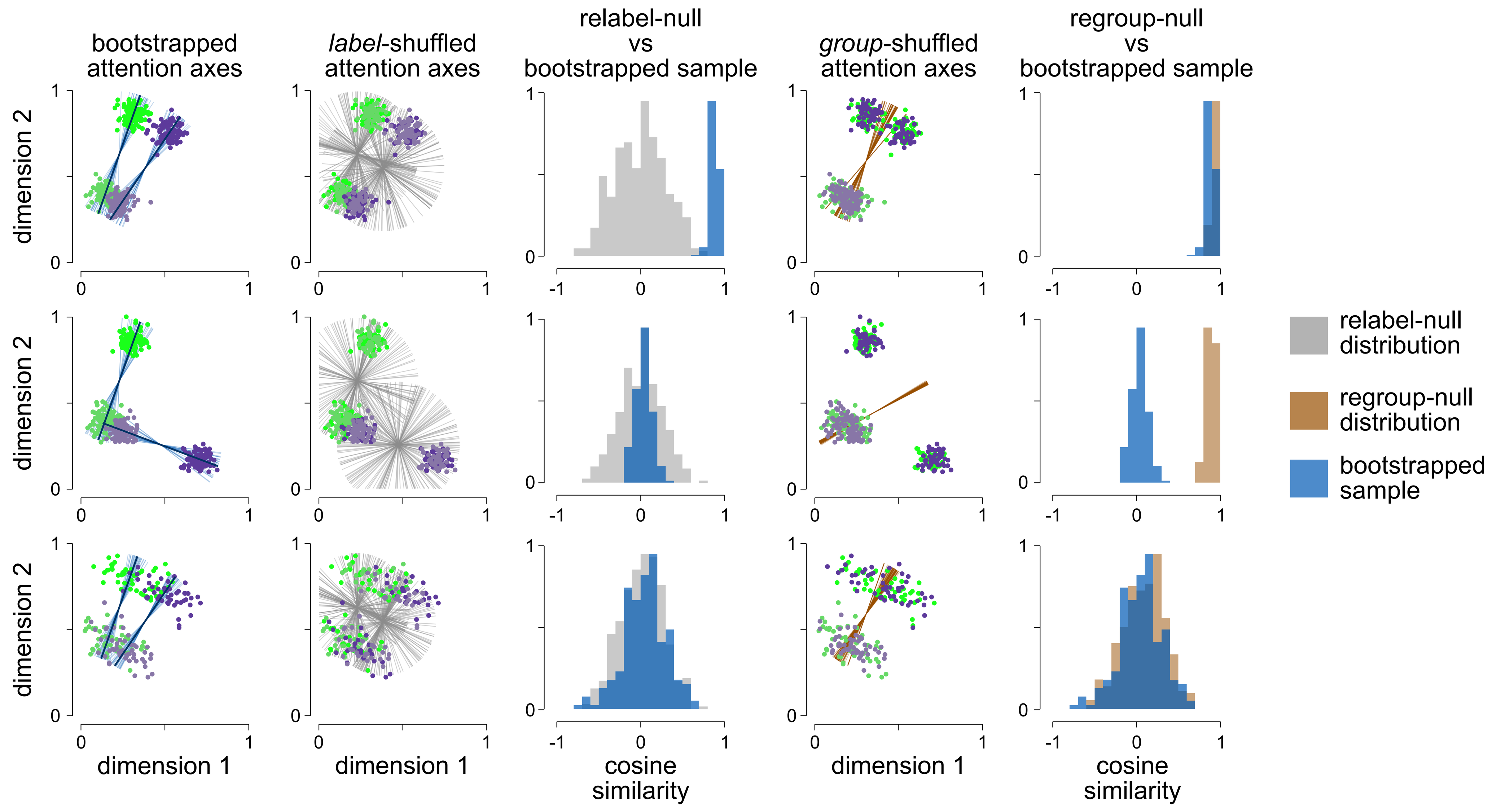
e. Preparatory states flexibly match task predictability



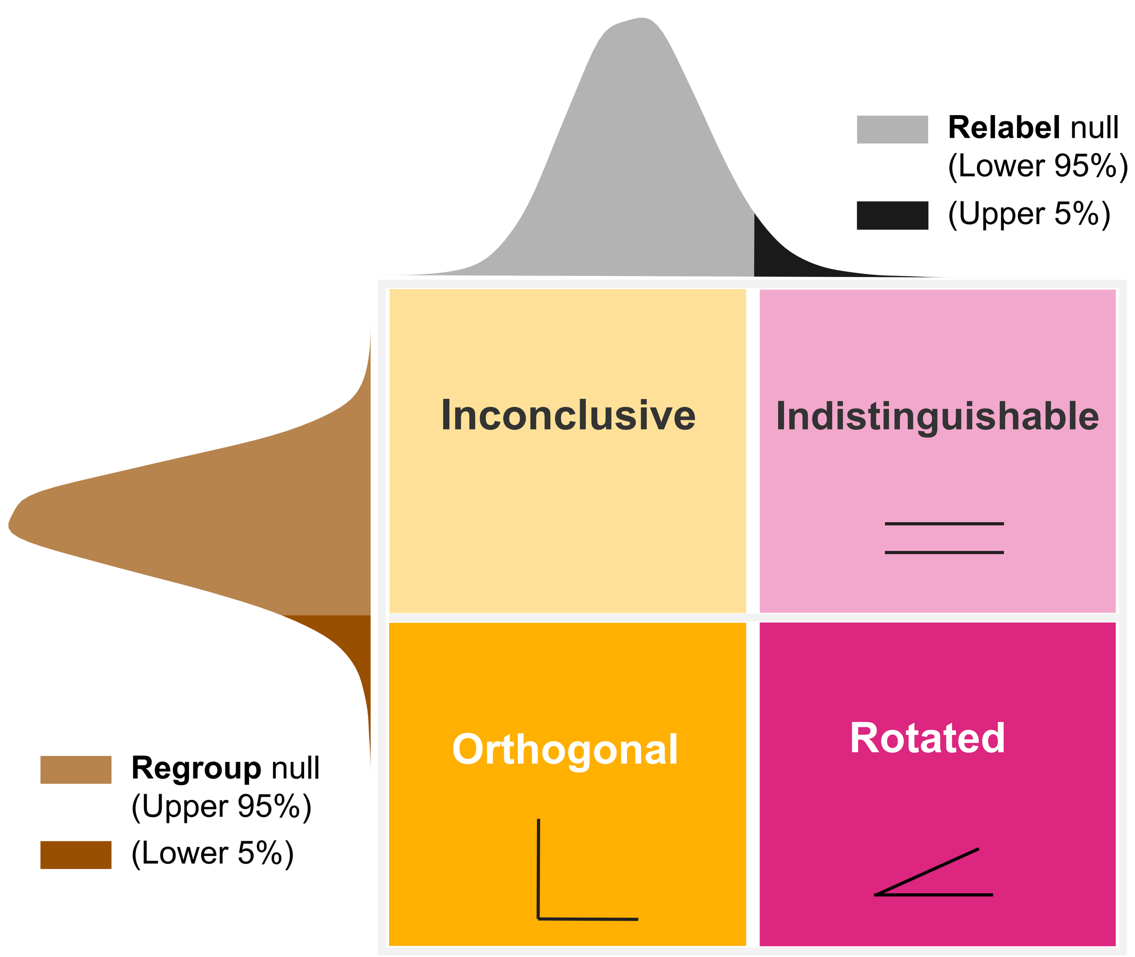
f. Color and contrast preparatory states occupy a shared subspace.



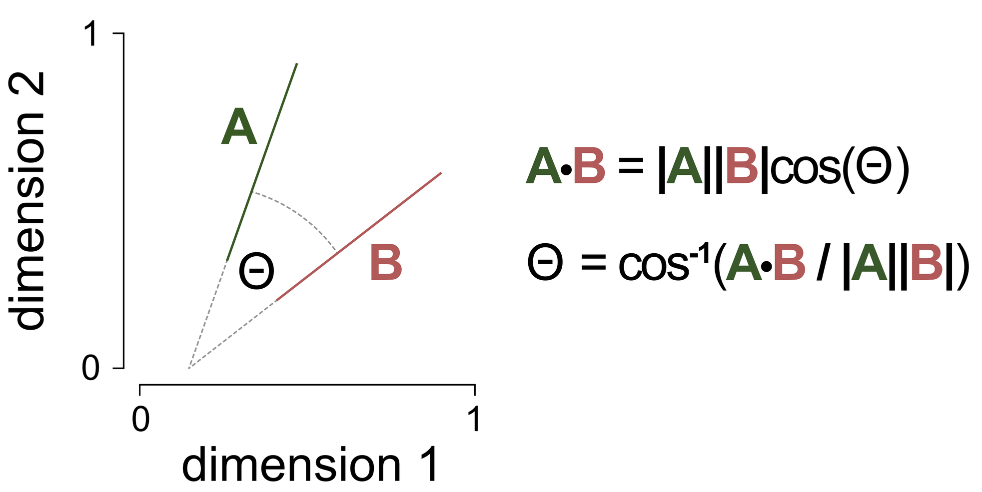
g. Two null distributions distinguish aligned, rotated, and orthogonal axes



h. Two null tests, four interpretive outcomes



i. Cosine similarity



j. Blocked sessions produce axis rotation and orthogonality.